

Dirty Spokes Dirtbike Rental Management System

CS340 Project Step 5 - Portfolio Submission

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Web App URL: <http://classwork.engr.oregonstate.edu:41801/>

Executive Summary

The Dirty Spokes Dirtbike Rental Management System began in Week 1 as a proposal for a database-driven web application to replace the paper-based rental tracking at a motocross facility in Southern California. The facility, operating 50 bikes and servicing approximately 500 customers across 2,500 annual sessions, required an efficient administrative solution. From that initial proposal to the final submission, the project underwent a series of iterative changes and upgrades driven by peer review and practical challenges during development.

The most significant structural change across the project's development was the refinement of the Rentals table's role and its foreign key behavior. In Step 1, Rentals was introduced as an intersection table that implemented the M:N relationship between Customers and Bikes with additional attributes such as rentalDate, returnDate, hourMeterOut, and hourMeterIn that gave it an independent, entity-level identity. Several peer reviewers flagged the initial Entity Relationship Diagram (ERD) for including a redundant connector between Customers and Bikes alongside the Rentals intersection. This was removed. Step 1 feedback also prompted the addition of explicit VARCHAR lengths, INT UNSIGNED designations, and cardinality markers to the ERD. All of these changes were accepted and carried forward.

The second major change involved ON DELETE behavior on the Rentals foreign keys (FK). The initial design used ON DELETE RESTRICT on both customerID and bikeID FKs, which was defended in Step 2 on the grounds of data-preservation. A TA and one peer reviewer challenged the ON DELETE RESTRICT behavior in Step 2, and the position was initially declined. However, during Step 3 implementation it became clear that RESTRICT made it impossible to demonstrate M:N deletion behavior as required by the project spec. This meant any customer with rental history could never be deleted without manually clearing those records first. Both FKs were updated to ON DELETE CASCADE in Step 3, satisfying the cascade requirement. This enabled clean parent-entity deletion through the intersection table.

On the implementation side, the project exceeded the minimum requirements. Rather than implementing CUD for a single entity, full CRUD via stored procedures was built for all 4 entities (Customers, Brands, Bikes, and Rentals). The DDL was wrapped in a reset_db() stored procedure invoked by a RESET button on every entity page. Minor UI fixes were made in response to Step 3 reviews, including enforcing consistent ORDER BY ASC on all browse queries and unifying button styling across the interface.

In terms of AI tooling, Claude (Anthropic) was used for web application scaffolding, UI template generation, and stored procedure routing patterns in Steps 3 and 4. This was effective in accelerating boilerplate generation for Handlebars templates, Express POST route structures, and stored procedure syntax. The primary strength of AI assistance was speed. Non-focus boilerplate that would have taken hours to write from scratch was generated in minutes. However, careful review was required and output sometimes needed schema-specific corrections where generated code defaulted to generic assumptions. All crucial database design decisions like entity modeling, normalization analysis, relationship direction, CASCADE behavior, and all accepted/declined feedback reasoning were worked through manually.

Project Outline

Dirty Spokes is a recreational dirtbike rental facility and track located in sunny Southern California, operating a fleet of 50 bikes across 8 major brands including Honda, Yamaha, and Kawasaki. The facility services roughly 500 unique customers per year across an estimated 2,500 annual rental sessions. Currently, the staff manage bike inventory, customer records, and rental history through manual bookkeeping and paper waivers.

Over the past year, this has led to a number of operational issues including customer records being lost or duplicated, no reliable way to track when a customer had which bike at any given time, and difficulty identifying when a bike was last rented or how many hours it has accumulated.

A database-driven web application allows Dirty Spokes administrators to manage Bikes, Brands, Customers, and Rentals in a unified system, ensuring accurate rental records, up-to-date usage metrics for bike maintenance and popularity, and smoother day-to-day operations at the track.

In terms of the database, the schema is in 3NF. All attributes contain atomic values with no repeating groups, all non-key attributes are fully dependent on their primary keys with no partial dependencies, and no non-key attribute determines another non-key attribute. The web application provides administrators with four dedicated management pages: Bikes, Brands, Customers, and Rentals. Each page supports full Create, Read, Update, and Delete operations through a browser-based interface. Detailed, in-depth explanations of all attributes and entity relationships can be found in the Database Outline on the next page.

Database Outline

Bikes: records the dirtbikes available in the rental fleet

- bikeID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- frameNumber: VARCHAR(50), unique, NOT NULL
- brandID: INT UNSIGNED, NOT NULL, FK → Brands.brandID
- modelName: VARCHAR(100), NOT NULL
- engineSize: VARCHAR(10), NOT NULL - e.g. "250cc", "450cc"
- bikeYear: INT UNSIGNED, NOT NULL
- engineHourMeter: DECIMAL(6,1), NOT NULL
- Relationships:
 - M:1 between Bikes and Brands, many bikes can belong to one brand implemented by FK brandID. ON DELETE RESTRICT prevents a brand from being deleted while bikes are associated with it. ON UPDATE CASCADE propagates brand ID changes to all associated bikes.
 - 1:M between Bikes and Rentals, one bike can appear in many rental records implemented by FK bikeID in Rentals
 - M:N between Bikes and Customers, a bike can be rented by many customers and a customer can rent many bikes, implemented through the Rentals intersection table

Brands: records the manufacturers of the bikes in the fleet

- brandID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- brandName: VARCHAR(100), unique, NOT NULL
- countryOfOrigin: VARCHAR(100), NOT NULL
- localDealer: VARCHAR(255), NOT NULL
- Relationships:
 - 1:M between Brands and Bikes, one brand can have many bikes implemented by brandID FK in Bikes

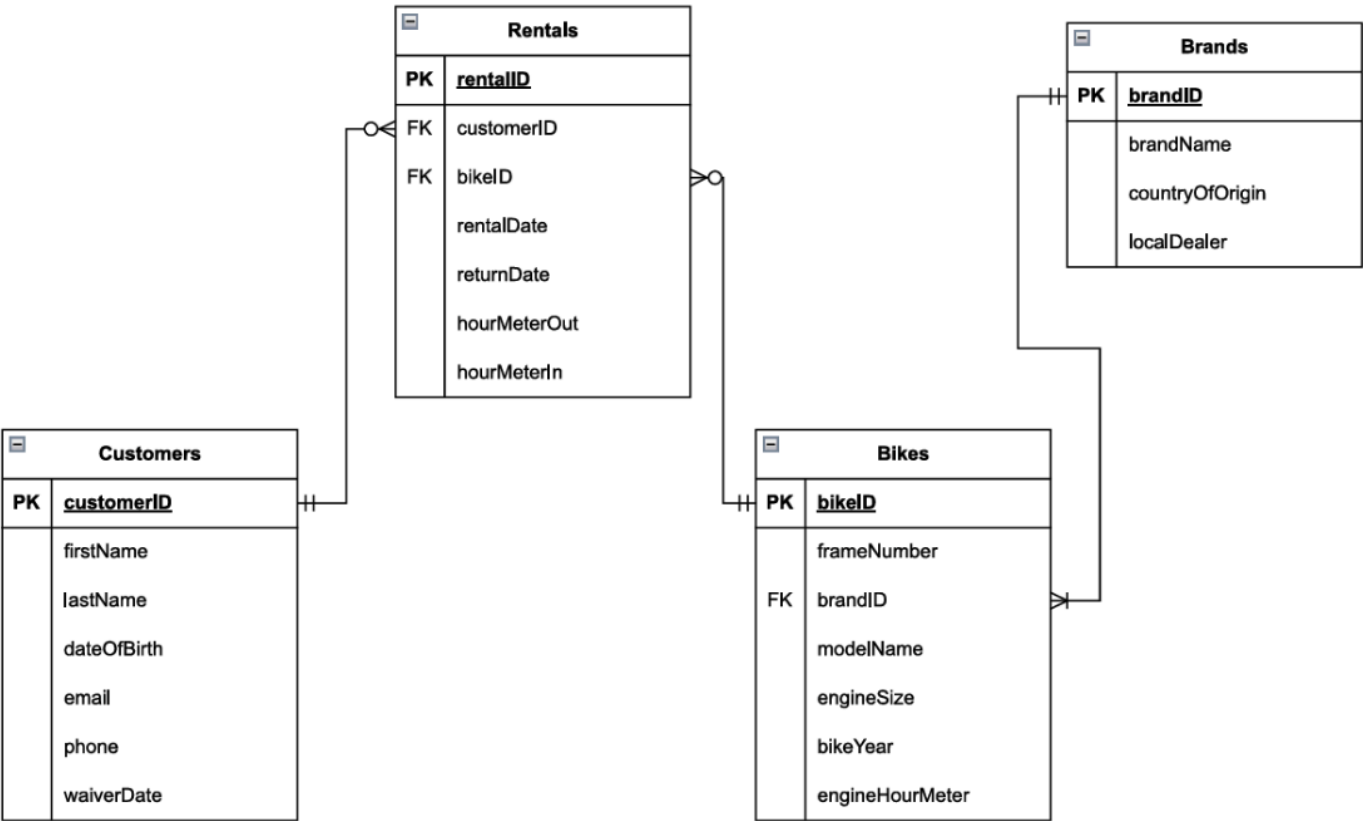
Customers: records customers who rent bikes at the track

- customerID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- firstName: VARCHAR(50), NOT NULL
- lastName: VARCHAR(50), NOT NULL
- dateOfBirth: date, NOT NULL
- email: VARCHAR(255), unique, NOT NULL
- phone: VARCHAR(20), NOT NULL
- waiverDate: date, NOT NULL
- Relationships:
 - 1:M between Customers and Rentals, one customer can have many rental records implemented by FK customerID in Rentals
 - M:N between Customers and Bikes, a customer can rent many bikes and a bike can be rented by many customers, implemented through the Rentals intersection table

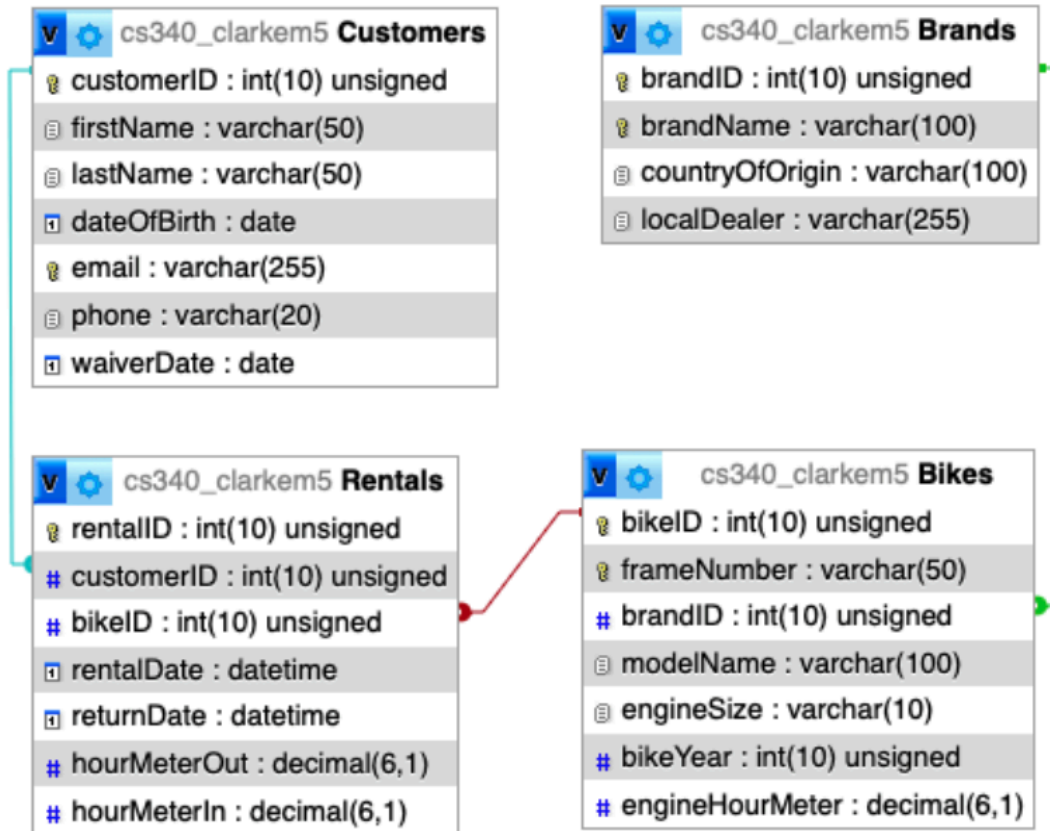
Rentals: intersection table recording which customer rented which bike and when

- rentalID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- customerID: INT UNSIGNED, NOT NULL, FK → Customers.customerID
- bikeID: INT UNSIGNED, NOT NULL, FK → Bikes.bikeID
- rentalDate: datetime, NOT NULL
- returnDate: datetime, NOT NULL
- hourMeterOut: DECIMAL(6,1), NOT NULL
- hourMeterIn: DECIMAL(6,1), NOT NULL
- Relationships:
 - M:1 between Rentals and Customers, many rental records can belong to one customer where Customers.customerID is PK and Rentals.customerID is FK. ON DELETE CASCADE automatically removes Rental records associated with a Customer when that Customer is deleted. ON UPDATE CASCADE propagates ID changes.
 - M:1 between Rentals and Bikes, many rental records can reference one bike where Bikes.bikeID is PK and Rentals.bikeID is FK. ON DELETE CASCADE automatically removes Rental records associated with a Bike when that Bike is deleted. ON UPDATE CASCADE propagates ID changes.
 - M:N between Customers and Bikes, a customer can rent many bikes and a bike can be rented by many customers

Entity-Relationship Diagram



Schema Diagram



Sample Data

Bikes Table

bikeID	frameNumber	brandID	modelName	engineSize	bikeYear	engineHourMeter
1	JYACB11C0RA015607	3	YZ65	65cc	2024	15.8
2	VBKXWM236MM321033	2	300 XC-W	293cc	2021	71.4
3	VNBS648C4PB000307	1	SE-F 500 Factory 4T	478cc	2023	225.5

Brands Table

brandID	brandName	countryOfOrigin	localDealer
1	Sherco	France	Trials Offroad
2	KTM	Austria	SoCal Moto
3	Yamaha	Japan	SoCal Moto

Customers Table

customerID	firstName	lastName	dateOfBirth	email	phone	waiverDate
1	Jeremy	Stevens	1989-06-11	jsteve0@email.com	555-0101	2024-11-22
2	Andrea	Hernandez	2003-11-25	andrea44@email.com	555-0202	2025-04-17
3	Roman	Lutoslawski	1978-08-26	roman444@email.com	555-0303	2020-08-20

Rentals Table

rentalID	customerID	bikeID	rentalDate	returnDate	hourMeterOut	hourMeterIn
1	3	3	2025-04-06 11:22:56	2025-04-08 16:43:16	219.9	225.5
2	3	2	2026-04-10 09:30:17	2026-04-13 10:17:44	60.6	68.0
3	2	2	2026-04-19 10:19:19	2026-04-19 15:22:04	68.1	71.4

UI Screenshots

Home / Index Page – Site Navigation

Dirty Spokes

[Home](#)[Customers](#)[Brands](#)[Bikes](#)[Rentals](#)

Dirty Spokes Dirtbike Rental Management System

Select a page below to manage records.

Page	Description
Customers	Browse, add, edit, and delete customer records
Brands	Browse, add, edit, and delete bike brands
Bikes	Browse, add, edit, and delete bikes in the rental fleet
Rentals	Browse, add, edit, and delete rental records

Customers Page – READ / CREATE / DELETE / RESET (Edit visible)

Dirty Spokes

[Home](#)[Customers](#)[Brands](#)[Bikes](#)[Rentals](#)

Customers

Browse Customers

ID	First Name	Last Name	Date of Birth	Email	Phone	Waiver Date	Actions
1	Jeremy	Stevens	1989-06-11	jsteve0@email.com	555-0101	2024-11-22	Edit Delete
2	Andrea	Hernandez	2003-11-25	andrea44@email.com	555-0202	2025-04-17	Edit Delete
3	Roman	Lutoslawski	1978-08-26	roman444@email.com	555-0303	2020-08-20	Edit Delete

Add New Customer

First Name

Last Name

Date of Birth

mm/dd/yyyy

Email

Phone

Waiver Date

mm/dd/yyyy

Add Customer

Admin Actions

RESET Database

Customers Page – EDIT / UPDATE

Dirty Spokes

HomeCustomersBrandsBikesRentals

Edit Customer

First Name

Jeremy

Last Name

Stevens

Date of Birth

06/11/1989

Email

jsteve0@email.com

Phone

555-0101

Waiver Date

11/22/2024

Update Customer

Cancel

Bikes Page – READ / CREATE / DELETE / RESET (Edit visible)

Dirty Spokes

HomeCustomersBrandsBikesRentals

Bikes

Browse Bikes

ID	Frame Number	Brand	Model	Engine Size	Year	Hour Meter	Actions
1	JYACB11C0RA015607	Yamaha	YZ65	65cc	2024	15.8	EditDelete
2	VBKXWM236MM321033	KTM	300 XC-W	293cc	2021	71.4	EditDelete
3	VNBS648C4PB000307	Sherco	SE-F 500 Factory 4T	478cc	2023	225.5	EditDelete

Add New Bike

Frame Number

Brand

-- Select Brand --

Model Name

Engine Size

e.g. 250cc

Year

Hour Meter

Add Bike

Admin Actions

RESET Database

Bikes Page – EDIT / UPDATE

Dirty Spokes

HomeCustomersBrandsBikesRentals

Edit Bike

Frame Number

JYACB11C0RA015607

Brand

Yamaha

Model Name

YZ65

Engine Size

65cc

Year

2024

Hour Meter

15.8

Update Bike

Cancel

Brands Page – READ / CREATE / DELETE / RESET (Edit visible)

Dirty Spokes

HomeCustomersBrandsBikesRentals

Brands

Browse Brands

ID	Brand Name	Country of Origin	Local Dealer	Actions
1	Sherco	France	Trials Offroad	EditDelete
2	KTM	Austria	SoCal Moto	EditDelete
3	Yamaha	Japan	SoCal Moto	EditDelete

Add New Brand

Brand Name

Country of Origin

Local Dealer

Add Brand

Admin Actions

RESET Database

Brands Page – EDIT / UPDATE

Dirty Spokes

HomeCustomersBrandsBikesRentals

Edit Brand

Brand Name

Sherco

Country of Origin

France

Local Dealer

Trials Offroad

Update Brand

Cancel

Rentals Page – READ / CREATE / DELETE M:N / RESET (Edit visible)

Dirty Spokes

HomeCustomersBrandsBikesRentals

Rentals

Browse Rentals

ID	Customer	Bike Model	Rental Date	Return Date	Hour Meter Out	Hour Meter In	Actions
1	Roman Lutoslawski	SE-F 500 Factory 4T	2025-04-06 18:22:56	2025-04-08 23:43:16	219.9	225.5	Edit Delete
2	Roman Lutoslawski	300 XC-W	2026-04-10 16:30:17	2026-04-13 17:17:44	60.6	68.0	Edit Delete
3	Andrea Hernandez	300 XC-W	2026-04-19 17:19:19	2026-04-19 22:22:04	68.1	71.4	Edit Delete

Add New Rental

Customer

-- Select Customer --

Bike

-- Select Bike --

Rental Date

mm/dd/yyyy, --:--:-- --

Return Date

mm/dd/yyyy, --:--:-- --

Hour Meter Out

Hour Meter In

Add Rental

Admin Actions

RESET Database

Rentals Page – EDIT / UPDATE M:N

Dirty Spokes [Home](#) [Customers](#) [Brands](#) [Bikes](#) [Rentals](#)

Edit Rental

Customer

Bike

Rental Date

Return Date

Hour Meter Out

Hour Meter In

Citations

The web application scaffolding and UI polish for Step 3, as well as the stored procedures and corresponding POST routes for Step 4 were written with the help of Claude (Anthropic). Prompts described our database schema and requested UI scaffolding templates for a Node.js/Express/Handlebars application, minor styling updates, and SQL/routing patterns consistent with the course-provided starter code. All prompts were written by Spencer Keating and Matthew Clarke. AI-generated code was reviewed, tested, and adapted to our schema by the team before integration.

All database design, entity modeling, relationship definitions, normalization analysis, DDL/DML/PL SQL authoring, decision-making regarding accepted and declined feedback, and all other work in this document were completed without the use of AI tools.